



Cyber Security Readiness in the Frontier AI Era

Frontier artificial intelligence (AI) technologies are the most advanced, cutting-edge AI models available and represent an anticipated step change in capability, enabling significantly more powerful automation, reasoning and decision making than previous generations of AI.

Cyber security is a national priority. Frontier AI presents a clear dual-use challenge – the same models that strengthen cyber defence can be exploited by malicious actors to conduct cyber activities and other forms of harm faster, cheaper and at greater scale. Frontier AI increases the risks posed by, and accelerates the consequences of, known vulnerabilities, legacy systems and weak cyber hygiene, creating a ‘vulnerability storm’ for entities.

Australian Government entities must ensure they are appropriately protected in the frontier AI era through the consistent application of cyber security fundamentals, which remain the Commonwealth’s most effective defence by strengthening its cyber security posture.

Policy Advisory

This Policy Advisory provides official whole-of-government policy advice for ensuring strong cyber security readiness in the frontier AI era.

Australian Government entities do not need access to the most advanced frontier AI models to stay protected. Effective cyber readiness is achieved through implementation of existing cyber security mitigations and practices consistent with existing technical advice published by the [Australian Signals Directorate](#) (ASD) and policy as outlined in the [Protective Security Policy Framework](#) respectively.

ASD has created the practical guidance [How to Act Now to Prepare for Frontier AI](#) - refer Annex A.

Commonwealth entities should treat the following actions as an immediate priority:

Review and assure compliance with the PSPF, ISM and Essential Eight
Confirm executive accountability for cyber security risk management for frontier AI
Engage with ASD and Home Affairs guidance on frontier AI
Identify and remediate material gaps that could be exploited by AI-enabled threat actors

Entities must also ensure the following principles and PSPF Requirements are followed:

Principle and PSPF Requirement	Details
Internet-facing Systems (Req. 0211)	Entities must develop a Technology Asset Stocktake and Technology Security Risk Management Plan for internet-facing systems within their technology estate, including

Principle and PSPF Requirement	Details
	cyber-security strategies such as restricting access to internal networks and applying network segregation and separation where required.
Secure Procurement and Supply Chains (Req. 0039 to 0049)	Procurement, contracts and third-party outsourced arrangements must contain proportionate security terms to avoid exposing the entity or the Australian Government to an unacceptable level of risk, including considering foreign ownership, control or influence (FOCI) risks.
Reduce Attack Surfaces and Paths (Req. 0105)	Entities must apply ASD's Essential Eight user application hardening mitigation strategy to Maturity Level Two, to significantly reduce the attack surface of users' workstations and devices.
Patch Applications and Operating Systems (Req. 0099 and 0100)	Entities must patch application and operating systems to Maturity Level Two under ASD's Essential Eight Maturity Model. Timely patching ensures that security vulnerabilities or other deficiencies are fixed, improving the usability or performance of an application or operating system. Refer to ASD's guidance Patching Applications and Operating systems for the recommended timeframes and prioritisations for applying patches.
Legacy Technologies (Req. 0093)	Legacy technologies often cannot be patched, as they're no longer supported by vendors. Entities must ensure ASD's guidance Managing the risks of legacy IT 's temporary mitigations for legacy technologies is implemented.
Zero Trust Principles (PSPF Table 24)	The Guiding Principles to Embed a Zero Trust Culture ensures that current and emergent risks stemming from a rapidly evolving cyber threat landscape and expansion of an entity's digital attack surface are appropriately managed through entity-wide transformation.
Gateways Security (Req. 0214)	Digital Infrastructure that processes, stores or communicates Australian Government security classified information must be protected by a Gateway or Security Service Edge in accordance with the Australian Government Gateway Security Standard. Gateways play a vital role in securing a technology system by providing entities with protection at its perimeter.
ASD's Cyber Security Partnership Program (Req. 0214 and 0215)	Entities must connect to ASD's Cyber Security Partnership Program and Cyber Threat Intelligence Sharing Platform to engage with the ASD's Australian Cyber Security Centre (ACSC) and fellow partners.
Apply the Information Security Manual (Req. 0084 and 0085)	Entities must apply ISM's cyber security principles are applied during all stages of the lifecycle of each system and that the ISM controls and cyber security guidelines are applied on a risk-based approach. The ISM outlines the controls to protect the entity's technology systems and data from cyber threats.

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Intended Audience Accountable Authorities
Chief Security Officers
Chief Information Security Officers
Procurement Officers
Entity Personnel

Contact Commonwealth Security Policy Branch, Department of Home Affairs
PSPF@homeaffairs.gov.au

FRONTIER AI: ORGANISATIONS MUST ACT NOW

Frontier AI is collapsing exploit timelines from days to hours. Lock down the fundamentals now.

N-days ➔ N-hours
Vulnerability Discovery to Exploitation

STRATEGY	01		02		03		04		05		06	
												
	Secure		Reduce		Replace		Prepare		Adopt		Modernise	
	<i>Horizon: Short-term</i>		<i>Horizon: Short-term</i>		<i>Horizon: Short-term</i>		<i>Horizon: Medium-term</i>		<i>Horizon: Medium-term</i>		<i>Horizon: Longer-term</i>	
	Systems are securely configured to approved and maintained baselines, including by reducing attack surfaces and attack paths, with configurations continually monitored and consistently enforced.		Vulnerabilities in systems are identified, documented, validated and prioritised for remediation or mitigation in a timely manner, with all remediation and mitigation actions verified for effectiveness.		Systems that are not capable of meeting cyber security requirements are managed using compensating controls, along with enhanced monitoring and assurance activities, to maintain an acceptable level of residual risk until they can be decommissioned or replaced.		Systems are securely configured to approved and maintained baselines, including by reducing attack surfaces and attack paths, with configurations continually monitored and consistently enforced.		Artificial intelligence is used for cyber defence and is secure, controllable, human-supervised and used in an ethical and accountable manner.		Systems are planned, designed, developed, tested, deployed, maintained and decommissioned according to business criticality ratings and security and resilience requirements using Secure by Design and Secure by Default principles and practices.	
	ACTIONS		ACTIONS		ACTIONS		ACTIONS		ACTIONS		ACTIONS	
	<i>Reduced attack surface</i> ■ Define and deploy approved configuration baselines. ■ Disable or remove insecure or unnecessary services, settings and communication protocols. ■ Detect and respond to configuration drift from approved configuration baselines through automation or continuous security assessments.		<i>Secure software vulnerabilities</i> ■ Identify software vulnerabilities and weaknesses through continuous security assessment activities. ■ Apply security patches, updates or vendor mitigation within risk-based timeframes or, where operational impact is low, automatically.		<i>Replace legacy information technology</i> ■ Establish and maintain a process for identifying, assessing and managing the cyber security risk associated with legacy systems. ■ Replace or isolate legacy systems where operationally possible.		<i>Prepare for cyber security incidents</i> ■ Review cyber security incident response, business continuity and disaster recovery plans to ensure they remain fit for purpose. ■ Regularly exercise cyber security incident response, business continuity and disaster recovery plans.		<i>Adopt AI for cyber defence purposes</i> ■ Deploy AI models for augmenting software development activities, such as identifying and remediating software flaws and weaknesses. ■ Deploy AI models for augmenting system assurance activities, such as performing vulnerability scanning and vulnerability assessments. ■ Deploy AI models for augmenting system monitoring activities, such as identifying and triaging cyber security events.		<i>Modernise for the future</i> ■ Design systems to enforce authorised access, and minimise default and required privileges. ■ Design systems to protect data. ■ Design systems to provide visibility.	

FRONTIER AI: ORGANISATIONS MUST ACT NOW

ACTIONS	DESCRIPTION	GUIDANCE
1. <i>Approved configuration baselines</i>	Define and deploy approved configuration baselines.	ISM controls: ISM-0912, ISM-1867, ISM-0687, ISM-0289, ISM-0290, ISM-1913, ISM-1858, ISM-1914, ISM-1409, ISM-1915, ISM-2109, ISM-1859, ISM-1860, ISM-1412, ISM-1916, ISM-1246. ACSC publications: Nil.
2. <i>Software configuration hardening</i>	Disable or remove insecure or unnecessary services, settings and communication protocols.	ISM controls: ISM-0383, ISM-0380, ISM-1654, ISM-1655, ISM-1621, ISM-1806, ISM-1470, ISM-1671, ISM-1260, ISM-1247, ISM-1603, ISM-1055, ISM-2076, ISM-2077, ISM-1304. ACSC publications: Hardening Microsoft Windows 11 workstations, Hardening Linux workstations and servers, Hardening Microsoft 365, Office 2021, Office 2019 and Office 2016.
3. <i>Approved configuration drift</i>	Detect and respond to configuration drift from approved configuration baselines through automation or continuous security assessments.	ISM controls: Nil. ACSC publications: Nil.
4. <i>Continuous security assessments</i>	Identify software vulnerabilities and weaknesses through continuous security assessment activities.	ISM controls: ISM-1807, ISM-1808, ISM-1698, ISM-1699, ISM-1700, ISM-1701, ISM-1702, ISM-1752, ISM-1703, ISM-1900. ACSC publications: Nil.
5. <i>Patching processes</i>	Apply security patches, updates or vendor mitigations within risk-based timeframes or, where operational impact is low, automatically.	ISM controls: ISM-1876, ISM-1690, ISM-1691, ISM-1692, ISM-1901, ISM-1693, ISM-1877, ISM-1694, ISM-1695, ISM-1696, ISM-1902, ISM-1878, ISM-1751, ISM-1879, ISM-1697, ISM-1903, ISM-1904, ISM-0300. ACSC publications: Patching applications and operating systems
6. <i>Legacy ICT management</i>	Establish and maintain a process for identifying, assessing and managing the cyber security risk associated with legacy systems.	ISM controls: ISM-1809. ACSC publications: Managing the risks of legacy IT: Executive guidance, Managing the risks of legacy IT: Practitioner guidance.
7. <i>Legacy ICT decommissioning</i>	Replace or isolate legacy systems where operationally possible.	ISM controls: ISM-1905, ISM-1704, ISM-0304, ISM-1501, ISM-1753, ISM-1981, ISM-1982. ACSC publications: Managing the risks of legacy IT: Executive guidance, Managing the risks of legacy IT: Practitioner guidance.
8. <i>Incident response planning</i>	Review cyber security incident response, business continuity and disaster recovery plans to ensure they remain fit for purpose.	ISM controls: ISM-0576, ISM-0888, ISM-0043. ACSC publications: Cyber security incident response planning: Executive guidance, Cyber security incident response planning: Practitioner guidance.
9. <i>Incident response exercising</i>	Regularly exercise cyber security incident response, business continuity and disaster recovery plans.	ISM controls: ISM-1784. ACSC publications: Cyber security incident response planning: Executive guidance, Cyber security incident response planning: Practitioner guidance.
10. <i>AI software test</i>	Deploy AI models for augmenting software development activities, such as identifying and remediating software flaws and weaknesses.	ISM controls: ISM-2122. ACSC publications: Opportunities for AI in cyber defence.
11. <i>AI security assessments</i>	Deploy AI models for augmenting system assurance activities, such as performing vulnerability scanning and vulnerability assessments.	ISM controls: ISM-2119. ACSC publications: Opportunities for AI in cyber defence
12. <i>AI security monitoring</i>	Deploy AI models for augmenting system monitoring activities, such as identifying and triaging cyber security events.	ISM controls: ISM-2117. ACSC publications: Opportunities for AI in cyber defence.
13. <i>System authentication security</i>	Design systems to enforce authorised access, and minimise default and required privileges.	ISM controls: ISM-0432, ISM-0434, ISM-0435, ISM-0405, ISM-1852, ISM-1507, ISM-1508, ISM-0445, ISM-0430, ISM-1591, ISM-1404, ISM-1648, ISM-1647, ISM-0407, ISM-1610, ISM-1611, ISM-1612. ACSC publications: Foundations for modern defensible architecture, Restricting administrative privileges.
14. <i>System data security</i>	Design systems to protect data.	ISM controls: ISM-1547, ISM-1548, ISM-1511, ISM-1810, ISM-1811, ISM-1812, ISM-1813, ISM-1705, ISM-1706, ISM-1814, ISM-1707, ISM-1708, ISM-1515. ACSC publications: Foundations for modern defensible architecture, Ten things to know about data security.
15. <i>System visibility provision</i>	Design systems to provide visibility.	ISM controls: ISM-1566, ISM-1509, ISM-1650, ISM-1613, ISM-1660, ISM-1889, ISM-1624, ISM-1976, ISM-1977, ISM-0582, ISM-1830, ISM-1978, ISM-1979, ISM-1683, ISM-1895, ISM-0580, ISM-1405, ISM-1983, ISM-1984, ISM-1985, ISM-1815, ISM-0988, ISM-0585, ISM-1959, ISM-1986, ISM-1906, ISM-1907, ISM-0109, ISM-1987, ISM-1960, ISM-1961, ISM-2116, ISM-1228, ISM-1988, ISM-1989, ISM-1910, ISM-2015, ISM-1536, ISM-1911, ISM-1537, ISM-1963, ISM-1964, ISM-0634, ISM-0670, ISM-0261. ACSC publications: Best practices for event logging and threat detection, Implementing SIEM and SOAR platforms: Executive guidance, Implementing SIEM and SOAR platforms: Practitioner guidance, Priority logs for SIEM ingestion: Practitioner guidance, Windows event logging and forwarding.